



YISHUN INNOVA JUNIOR COLLEGE
JC 1 PROMOTIONAL EXAMINATION
Higher 2

CANDIDATE
NAME

CG

INDEX NO

COMPUTING

Paper 1 Written

9569/01

26 Sep 2024

1 hour 10 minutes

Additional Materials: 4-page Answer booklet

READ THESE INSTRUCTIONS FIRST

An answer booklet will be provided with the question paper.

You should follow the instructions on the front cover of the answer booklet.

If you need additional answer paper ask the invigilator for a continuation booklet.

Answer **all** questions.

Approved calculators are allowed.

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is **40**.

1

Index	0	1	2	3	4	5	6	7
arr	33	37	42	51	62	71	80	83

- (a) Describe using the linear search algorithm to find the value 42 in the array `arr`. [2]
- (b) Describe using the binary search algorithm to find the value 42 in the array `arr`. [3]
- (c) A student claims that both the linear search and binary search algorithms are equally efficient because they require the same number of steps to find the value 42.
Explain whether the claim is true. [4]

2 The following is an unsorted array:

`array = [5, 2, 7, 1, 3, 8, 6, 4]`

- (a) (i) With the aid of a diagram, describe how the elements can be arranged in ascending order using the merge sort algorithm. [4]
- (ii) State the order of growth for the merge sort algorithm. [1]

The elements in the unsorted array are inserted into a binary search tree (BST) in the order of the index number.

- (b) (i) Draw the resulting BST. [3]
- (ii) Describe how the elements could be retrieved in an ascending order from the BST. [2]

- 3** RGB (red, green and blue) is a system of representing the colours for digital lighting devices. Red, green and blue can be combined in various proportions to obtain any colour in the visible spectrum.

In the following table, a colour is represented with two hexadecimal digits each to represent the red, green and blue components, and each colour component has a value ranging from 00 to FF:

Colour	RGB Colour (Hexadecimal Representation)
Black	00:00:00
White	FF:FF:FF
Red	FF:00:00
Green	00:FF:00
Blue	00:00:FF
Violet	EE:82:EE
SeaGreen	2E:8B:57
Chocolate	D2:69:1E
MidnightBlue	19:19:70

- (a) With clear working, show the steps required to convert the hexadecimal representation for the `SeaGreen` colour to a decimal representation. [2]

A student is writing a program to remotely control the LED lightings on a drone. The communication requires the data to be in binary numbers. Three sets of 8-bit binary data will be used to represent the red, green and blue colour components.

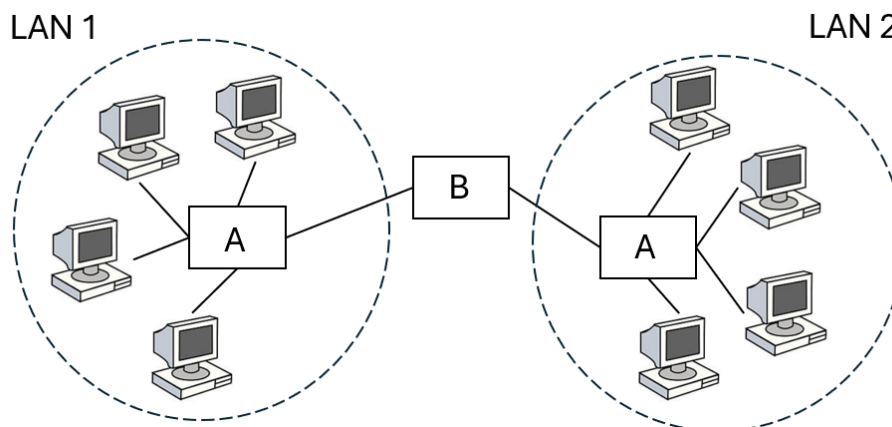
- (b) With clear working, show the steps required to convert the hexadecimal representation for the `Chocolate` colour to a binary representation. [2]
- (c) State and explain two advantages of using the hexadecimal representation for numbers in computer systems, particularly when working with digital data, as compared to using the decimal representation. [2]

- 4 The following database table `OrderDetails` is used to track customers' purchases:

CustomerID	ProductID	Quantity	UnitPrice	Date
C001	P2001	2	10.00	2024-09-10
C001	P2002	1	15.00	2024-09-10
C001	P2003	3	7.50	2024-09-11
C002	P2001	1	10.00	2024-09-12
C002	P2004	5	8.00	2024-09-12
C003	P2002	2	15.00	2024-09-13
C003	P2005	1	20.00	2024-09-13
C004	P2003	4	7.50	2024-09-14
C004	P2003	1	7.50	2024-09-15

- (a) Suggest a suitable Primary Key (PK) for the table `OrderDetails`. [1]
- (b) Explain two functions of a PK for a table in a relational database. [2]
- (c) (i) Suggest a modification to the table so that a single attribute can be used as the PK. [1]
- (ii) State a new PK with a single attribute and explain how this is better than the original PK suggested in **part (a)**. [2]

- 5 Two companies connected their local area network (LAN) using network cables and devices.



- (a) Name the network devices A and B used to connect the computers within a LAN and across the LANs. [2]
- (b) State two main differences between the devices A and B. [2]
- (c) State one advantage and one disadvantage of connecting networks together using cables. [2]

Instead of connecting the networks directly using cables, they can be connected via the internet.

- (d) State and explain one disadvantage of connecting via the internet. [2]
- (e) Suggest a solution for the concern stated in **part (d)**. [1]

END OF PAPER

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